

02YMECH - Solution 7

Assigned for the week of Nov 3, 2025

Solutions

$$r(\theta) = \frac{a}{1 + b \cos \theta}, \quad a > 0, \ 0 < b < 1.$$

As we derived in lectures:

$$f(r) = -\frac{l^2 u^2}{m} \left(\frac{d^2 u}{d\theta^2} + u \right),$$

where $u = 1/r$. Then

$$u = \frac{1}{a}(1 + b \cos \theta), \quad \frac{d^2 u}{d\theta^2} + u = \frac{1}{a}.$$

1. Central force

$$f(r) = -\frac{l^2}{ma} \frac{1}{r^2}.$$

Thus the force is an inverse-square law.

2. Attractive or repulsive

Since $m > 0$, $l^2 > 0$, and $a > 0$,

$$f(r) < 0 \quad \text{for all } r > 0,$$

so the force is *attractive*, in particular for $0 < r < a$.